National annex NC (informative)

Sampling and testing for acceptance inspection at delivery

NC.1 For acceptance at delivery, when requested, a spot sample of the cement should be taken in accordance with 3.6 and 6.2, 6.3, 6.4 or 6.5 of BS EN 196-7:1992 either before or at the time of delivery. A laboratory sample should be prepared and packed in accordance with clauses 8 and 9 of BS EN 196-7:1992. A sampling report should be completed at the time of sampling and should be attached to the laboratory sample in accordance with clause 10 of BS EN 196-7:1992.

NOTE: Testing may be delayed for up to three months from the time of sampling provided that there is confirmation that the sample has been stored continuously in the manner described in 9.2 of BS EN 196-7:1992.

NC.2 When the cement is tested for strength (see 7.1), it is recommended that the pit/quarry from which the CEN Standard sand (see BS EN 196-1) is obtained and the compaction procedure to be used should be those in use by the manufacturer at the time the cement was originally tested.

NOTE: It should be noted that the source of CEN Standard sand and the compaction procedure can, within permitted limits (see BS EN 196-1), influence the strength achieved.

NC.3 When the cement is tested for chemical properties (see 7.3) the test sample should be prepared by the method described in clause 6 of BS EN 196-2:1995.

NC.4 Testing should be carried out in accordance with the relevant methods in the BS EN 196 series of standards.

NC.5 The limiting values applicable to acceptance inspection of cement should be those given in Table NC.1.

NOTE: The acceptance inspection limits are in general those given as limit values for single results in Table 8 of this standard. However, Table 8 does not give values for loss on ignition or insoluble residue.

Table NC.1 Acceptance inspection limits

<table>
<tr><th colspan="2" rowspan="2">Property</th><th colspan="6">Strength class</th></tr>
<tr><th>32,5N</th><th>32,5R</th><th>42,5N</th><th>42,5R</th><th>52,5N</th><th>52,5R</th></tr>
<tr><td rowspan="2">Early strength (MPa)
lower limit value</td><td>2 day</td><td>—</td><td>8,0</td><td>8,0</td><td>18,0</td><td>18,0</td><td>28,0</td></tr>
<tr><td>7 day</td><td>14,0</td><td>—</td><td>—</td><td>—</td><td>—</td><td>—</td></tr>
<tr><td>Standard strength (MPa)
lower limit value</td><td>28 day</td><td>30,0</td><td>30,0</td><td>40,0</td><td>40,0</td><td>50,0</td><td>50,0</td></tr>
<tr><td colspan="2">Initial setting time (min)
lower limit value</td><td colspan="2">60</td><td colspan="2">50</td><td colspan="2">40</td></tr>
<tr><td colspan="2">Soundness (mm)
upper limit value</td><td colspan="6">10</td></tr>
<tr><td rowspan="3">Sulfate content
(as % SO_3 by mass)
upper limit value</td><td>CEM I
CEM II
CEM IV
CEM V</td><td colspan="3">4,0</td><td colspan="3">4,5</td></tr>
<tr><td>CEM III/A
CEM III/B</td><td colspan="6">4,5</td></tr>
<tr><td>CEM III/C</td><td colspan="6">5,0</td></tr>
<tr><td colspan="2">Chloride content (% by mass)[1)]
upper limit value</td><td colspan="6">0,10</td></tr>
<tr><td colspan="2">Loss on ignition (% by mass)
upper limit value</td><td colspan="6">5,1</td></tr>
<tr><td colspan="2">Insoluble residue (% by mass)
upper limit value</td><td colspan="6">5,1</td></tr>
<tr><td colspan="2">Pozzolanicity</td><td colspan="6">positive at 15 days</td></tr>
<tr><td colspan="8">[1)] Cement type CEM III may contain more than 0,10 % chloride but in such a case it is necessary to declare the actual chloride content.</td></tr>
</table>

National annex ND (informative)

Special Portland cements

ND.1 Controlled fineness Portland cement

Controlled fineness Portland cement should conform to the requirements for type CEM I of this British Standard and, in addition its fineness should lie within a range to be agreed between the purchaser and the manufacturer.

NOTE: For many years there has been a demand by specialist users for a cement which makes it easier to remove excess water from the concrete during compaction. In some applications the fineness of the cement is more critical than its compressive strength.

ND.2 Pigmented Portland cement

Pigmented Portland cement should be deemed to conform to the requirements for type CEM I of this British Standard provided that:

a) the pigments conform to BS EN 12878;

b) the chemical properties of the Portland cement constituent, excluding pigment conform to the requirements of clause 7.3 of this standard;

c) the final cement, including pigment, conforms to this British Standard with the exception of clause 6 and clause 7.3.

NOTE: The quantity of pigmented cement used in a concrete or mortar mix may need to be greater than that of an unpigmented cement in order to take account of the amount of pigment present.

National annex NE (normative)

Requirement, in the UK, for the loss on ignition of a siliceous fly ash constituent

In the UK, the loss on ignition of a siliceous fly ash constituent shall not exceed 7,0 % by mass, as a characteristic value.

NOTE 1: In 5.2.4.1 the requirement for the loss on ignition property of a siliceous fly ash constituent is permitted, within stated limits, to be standardized on a national basis.

NOTE 2: See 5.2.4.1 where the additional requirements for durability, compatibility with admixtures and the declaration to be made on the packaging or delivery note for the cement are given, where a requirement for the loss on ignition of a siliceous fly ash constituent has been standardized nationally.

National annex NF (informative)

Product guidance

NF.1 General

Guidance on the use of cements in concrete can be found in BS 5328-1 and in BS 8000-2.

Guidance on their use in mortar can be found in BS 5262, BS 5628-3 and BS 8000-3.

NF.2 Safety warning

NF.2.1 Manual handling of bags

Manual handling activities are subject to the Manual Handling Operations Regulations 1992 [1]. Where manual handling operations cannot be avoided, the Regulations require that the risks be assessed and reduced so far as is reasonably practicable. Guidance on how to assess and reduce risk, is given by the Health and Safety Executive (HSE), the UK's regulatory authority, in its booklet, Manual Handling (Manual Handling Operations Regulations 1992) [2], Guidance and Regulations L23 (HMSO). In addition, the HSE in its Construction Information Sheet No. 26 (Revised 10/96) [3], specifically encourages the use of 25 kg bags of cement whilst discouraging the use of 50 kg bags, in order to reduce the risk of injury.

NF.2.2 Safety in use

NF.2.2.1 Regulations

Work with cement is subject to the Control of Substances Hazardous to Health Regulations (COSHH) 1999 [4]. In addition, Portland cement has been classified as an irritant under The Chemicals [Hazard Information & Packaging] Regulations (CHIP)1994 [5].

These regulations variously require that:

a) the health risks of the cement in use be assessed and then prevented or controlled;

b) product health and safety information sheets be made available from the manufacturer/supplier;

c) bags containing common cement be labelled with a health and safety warning indicating that cement is an irritant.

NF.2.2.2 Hazards

When cement is mixed with water, for example when making concrete or mortar, or when cement becomes damp, a concentrated alkaline solution is produced. Where this comes into contact with the eyes or skin it may cause serious burns and ulceration. The eyes are particularly vulnerable and injury will increase with contact time.

Concentrated alkaline solutions in contact with skin tend to damage the nerve endings first before damaging the skin. Chemical burns can develop without pain being felt at the time.

In addition, cementitious grouts, cement-mortar and concrete mixes may, until they have set, cause both irritant and allergic contact dermatitis:

a) Irritant contact dermatitis results from a combination of the moisture content, alkalinity and abrasiveness of the construction materials.

b) Allergic contact dermatitis is mainly a consequence of the sensitivity of an individual's skin to hexavalent chromium salts in solution.

High repeated exposures to airborne cement in excess of the Occupational Exposure Standard (OES) [6]have been linked with rhinitis and coughing.

NF.2.2.3 First aid measures

a) In the event of eye contact, wash eyes immediately with copious amounts of clean water for a period of at least fifteen minutes and seek medical advice without delay;

b) In the event of skin contact, wash the affected area thoroughly with soap and water before continuing the activity. If irritation, pain or skin trouble occurs, seek medical advice;

c) In the event of ingestion, do not induce vomiting but wash out the mouth with water and give plenty of water to drink. If pain occurs, seek medical advice.

Clothing or footwear contaminated by wet cement, cementitious grout, cement-mortar or concrete should be removed and washed immediately and thoroughly before being re-used.

NF.2.2.4 Use of personal protective equipment (PPE)

a) Where the risk of cement becoming airborne can neither be prevented nor completely controlled, appropriate respiratory protective equipment should be worn to ensure that exposure is less than the regulatory limit [Occupational Exposure Standard (OES)]; and, in addition, dust-proof goggles should be worn in order to protect the eyes;

b) Where the risks from contact with wet cement or wet cement-containing construction materials can neither be prevented nor completely controlled, appropriate protective equipment should be worn as follows:

   1) *Protective clothing* should be worn in order that cement, or any cement/water mixture, e.g. concrete or mortar, does not come into contact with the skin. In some circumstances, such as when laying concrete, waterproof trousers and wellington boots may be necessary. Particular care should be taken to ensure that wet concrete does not enter the boots and that individuals do not kneel on wet concrete. Should wet concrete (mortar or grout) enter boots, gloves or other protective clothing, then the item(s) of clothing should be removed immediately and the skin thoroughly washed with soap and water. Items of clothing should be washed before re-use.

   2) Where this takes the form of *eye protection*, wherever there is a risk of cement, or any wet cement mixture entering the eye, dust-proof goggles should be worn.

NF.3 Storage

To protect cement from premature hydration after delivery, bulk silos should be waterproof and internal condensation should be minimized.

Paper bags should be stored clear of the ground, not more than eight bags high and protected by a waterproof structure. As significant strength losses begin after 4 weeks to 6 weeks of storage in bags in normal conditions, and considerably sooner under adverse weather conditions or high humidity, deliveries should be controlled and used in order of receipt. Manufacturers are able to provide a system of marking a high proportion of the bags in each delivery to indicate when they were filled.

NF.4 Test temperature

BS EN 196-1 and BS EN 196-3 require that the strength and setting time tests are carried out at a temperature of (20 ± 1) °C. When cement is tested at a different temperature the results are likely to be affected. Appropriate advice may be obtained from the manufacturer.

NF.5 Grouting and rendering

Where cement is to be used in grouts or renders that are pumped through small apertures, such as spray nozzles, it is recommended that the user passes the cement or suspension through a screen of suitable mesh aperture to retain any occasional coarse particles.

NF.6 Heat generation

The cement hydration process generates heat, particularly in the first few days. Cements with higher early strength usually have a higher initial rate of heat generation than those with lower early strength. A higher initial rate of heat generation may be an advantage for thinner concrete sections in cold weather because it reduces the need for extended striking times and the tendency for early-age frost damage. Conversely, it may be a disadvantage for larger concrete sections in either hot or cold weather on account of the temperature gradients which are set up.

NF.7 Alkali-silica reaction

Portland slag cements (CEM II/A-S and II/B-S), blastfurnace cements (CEM III/A, III/B and III/C), Portland fly ash cements (CEM II/A-V and II/B-V) and Pozzolanic (siliceous fly ash) cements (CEM IV/A and IV/B) may be beneficial in minimizing the risk of damage to concrete caused by the alkali-silica reaction.

NF.8 Sulfate resistance (including the thaumasite form)

Blastfurnace cements (CEM III/B and III/C), Portland fly ash cement (CEM II/B-V) and Pozzolanic (siliceous fly ash) cements (CEM IV/A and IV/B) can confer concrete with sulfate-resisting properties (see BS 5328-1).

Portland limestone cements, CEM II/A-L and CEM II/A-LL, for use in concrete exposed to a sulfate-bearing environment, are recommended in Class 1 sulfate conditions only, as defined in BS 5328-1.

National annex NG (informative)

Publications referred to in national annexes

NG.1 Standards publications

BS 12:1996, *Specification for Portland cement*

BS 146:1996, *Specification for Portland blastfurnace cements*

BS 915-2:1972, *Specification for high alumina cement – Part 2: Metric units*

BS 1370:1979, *Specification for low heat Portland cement*

BS 4027:1996, *Specification for sulfate-resisting Portland cement*

BS 4246:1996, *Specification for high slag blastfurnace cement*

BS 4248:1974, *Specification for supersulfated cement*

BS 5262:1991, *Code of practice for external renderings*

BS 5328-1:1997, *Concrete – Part 1: Guide to specifying concrete*

BS 5628-3:1985, *Code of practice for use of masonry – Part 3: Materials and components, design and workmanship*

BS 6588:1996, *Specification for Portland pulverized-fuel ash cements*

BS 6610:1996, *Specification for Pozzolanic pulverized-fuel ash cement*

BS 7583:1996, *Specification for Portland limestone cement*

BS 8000-2.1:1990, *Workmanship on building sites – Part 2: Code of practice for concrete work – Section 2.1: Mixing and transporting concrete*

BS 8000-2.2:1990, *Workmanship on building sites – Part 2: Code of practice for concrete work – Section 2.2: Sitework with in situ and precast work*

BS 8000-3:1989, *Workmanship on building sites – Part 3: Code of practice for masonry*

BS 8110-1:1997, *Structural use of concrete – Part 1: Code of practice for design and construction*

BS EN 196-1:1995, *Methods of testing cement – Part 1: Determination of strength*

BS EN 196-2:1995, *Methods of testing cement – Part 2: Chemical analysis of cement*

BS EN 196-3:1995, *Methods of testing cement – Part 3: Determination of setting time and soundness*

BS EN 196-7:1992, *Methods of testing cement – Part 7: Methods of taking and preparing samples of cement*

BS EN 197-2:2000, *Cement – Part 2: Conformity evaluation*

BS EN 12878:1999, *Pigments for colouring building materials based on cement and/or lime – Specifications and methods of test*

NG.2 Other publications

[1] GREAT BRITAIN. Manual Handling Operations Regulations 1992. London: The Stationery Office.

[2] GREAT BRITAIN. Manual Handling Operations Regulations 1992, Guidance and Regulations booklet L23. London: The Stationery Office.

[3] GREAT BRITAIN. Health and Safety Executive. Construction information sheet No. 26, 1996. London: HSE books.

[4] GREAT BRITAIN. Control of Substances Hazardous to Health Regulations (COSHH) 1999. London: The Stationery Office.

[5] GREAT BRITAIN. The Chemicals [Hazard Information and Packaging] Regulations (CHIP) 1994. London: The Stationery Office.

[6] GREAT BRITAIN. Occupational Exposure Criteria Document, Portland Cement Dust. London: HSE books.

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